

Part II . Interactive Data Visualization (50p).

The assignment will test the student's ability to read and process datasets, visualize data with *Ipywidgets*, *Plotly*, or other *interactive visualization libraries*, analyze data using visualization techniques, and select an appropriate data representation based on the type of data and/or the question being asked.

CT2. Interactive Visualization Based on Case A of Part I (25p)

The dataset you will work with is the one presented in Case A of Part I. Based on it, you are required to add categorical selection and time-window selection to your visualizations.

CT2-1. Categorical selection

You have to create an interactive scatter plot that lets users select one or more companies to visualize their stock price trends over time. **Requirements:**

- (1) Use “date” as the x-axis and “stock price” as the y-axis.
- (2) Ensure the chart design follows the guidelines and principles of effective visualizations discussed during the lectures, e.g. clear axis labels, a legend, and color differentiation for companies.
- (3) You can use *plotly.express*, *plotly.graph_objects*, or *Matplotlib* and *IpyWidgets* to create the plot.
- (4) Select the two stocks of TSLA and AMZN in the visualization and identify the day when the stock price of TSLA is higher than AMZN. Answer the time and stock price details in text.

CT2-2. Time window selection.

Implement an interactive feature of selection of a time window/range in the chart. Add a representation of the selected time window to the visualization to customize the selected time window and calculate the average stock price for that range. Moving the start and end times updates the selected time window in the visualization. Requirements:

- (1) Visualize the customizable time window on your visualization created in CT2-1.
- (2) Consider that the time window control should not affect the categorical selection in CT2-1.
- (3) Ensure the chart design follows the guidelines and principles of effective visualizations discussed during the lectures, e.g. clear axis labels, a legend, and color differentiation for companies.
- (4) Analyze: during July 1st-August.30th 2024, what is the average stock price of Company AAPL? Answer in text with explanations based on interactive visualization.

(Hints: key features for this visualization could be similar to the Figure 1 below, but not limited to.)



Figure 1. Key features for this visualization (CT2-1).

Case C: Interactive visualization of geographical data

You are tasked with performing a geographical analysis of city populations worldwide. A dataset of the top 3 cities with the largest populations in each country is provided ("assignment_CaseC_cities_populations.csv"). The dataset contains city names, the countries where each city is located, the city's population, and the city's position (latitude and longitude). Now you are supposed to create interactive visualizations and identify data patterns from them.

Task 3: Interactive visualization of population on a map (8p)

Based on the provided dataset, create an interactive geographical visualization that presents a population overview and allows exploration of the area of interest. **Requirements:**

- (1) Load the provided CSV file into a pandas DataFrame. Inspect the dataset to understand its structure and identify missing information (e.g., missing population value). Address the missing information in each row and elaborate on your solution in text.
- (2) Create an interactive geographical scatter plot showing city locations, with marker size based on city population, and marker color based on population density.
(Hints: Use `plotly.express.scatter_geo` or similar tools for visualization.)
- (3) Based on the interactive visualization created, analyze in text the population patterns in European countries in terms of the overall range, and cities with the highest population density.

Task 4: Interactive Dashboard (17p)

T4-1. Interactive bar chart of top N cities by population density (8p)

Based on the provided dataset, create another interactive bar chart that represents the population data of the top N cities with the largest populations, and analyze the top 10 cities (N=10) with the largest populations from your interactive visualization. **Requirements:**

- (1) Create an interactive bar chart of the top N cities with the highest populations using `Ipywidgets` in Jupyter Notebook. N is the value of the `IntText` widget for dynamically selecting the number of top cities.
- (2) Identify the top 10 (N =10) cities with the largest population from the interactive visualization. Explore the locations of these cities in the geographic scatter plot above and elaborate on the geographical patterns in text.

T4-2. Interactive Mini-Dashboard (Linking and Brushing) (9p)

Based on the two interactive visualizations you created above, build a mini dashboard to explore the relationship between cities' geographic locations and their populations. You need to implement linking and brushing to allow interactive data exploration across multiple views. **Requirements:**

- (3) Visualize the interactive geographic visualization and interaction bar chart together as two views (`View_1` and `View_2`). Create a selection tool (such as a bounding box, lasso, or interval selection) on at least one of your views.
- (4) When a user "brushes" or selects a group of data points in `View_1`, `View_2` must dynamically update to reflect only the selected data (filtering) or highlight the selected data while dimming the rest (highlighting).

A visual illustration of the required interactive features for the mini dashboard is shown in Figure 2 below. (Hints*: you may use libraries like Altair, Plotly Dash, Bokeh for creating the mini dashboard.)

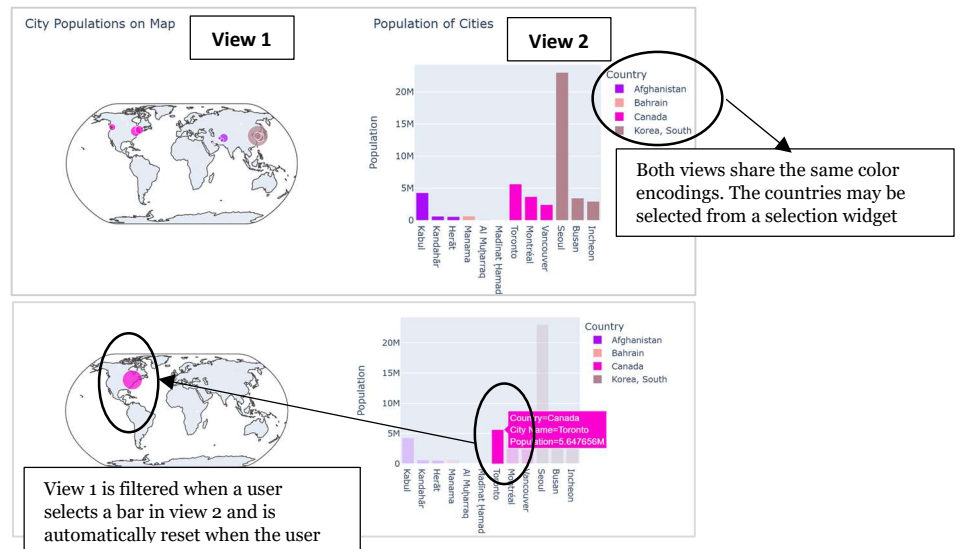


Figure 2. A visual illustration of key interactive features in the mini dashboard (T4-2).